OpenCloning: Open Source DNA engineering that integrates with your Lab Software

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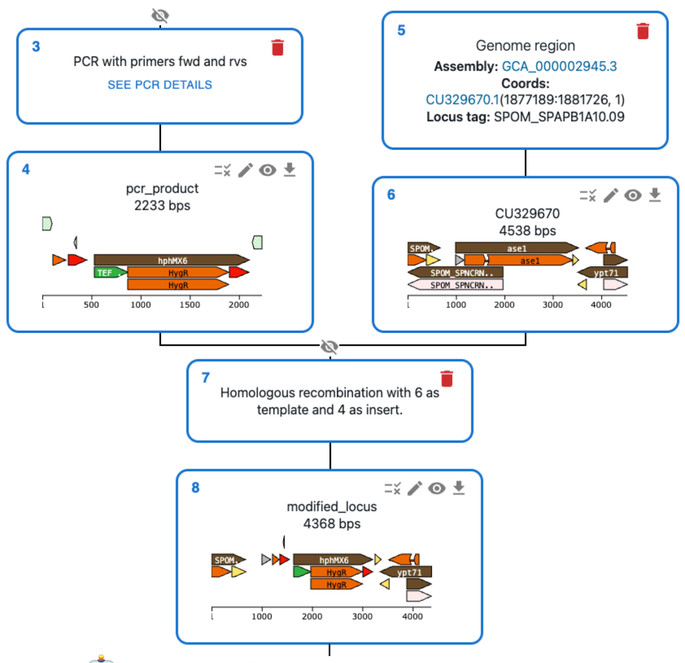
1. University College London, London, U.K. 2. The Francis Crick Institute, London, U.K.

What is OpenCloning?

OpenCloning is a set of tools for designing and simulating DNA engineering workflows.

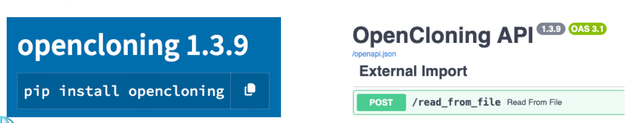
Web-based workflow editor

State of the art interface for assembly, annotation and primer design.



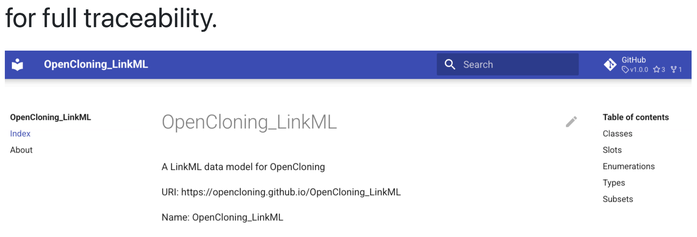
Automation via python or web API

All design steps can be automated via python or web API.



Data model for DNA engineering workflows

A well defined data model for DNA engineering workflows for full traceability.



Are you a LIMS/ELN vendor?

Integrate OpenCloning into your platform

- Unlock **new revenue streams** by offering a **complete DNA engineering solution** to your customers.
- Teams working with DNA will default to platforms with DNA engineering capabilities, Benchling and Dotmatics primarily .
- If you **bring those DNA design capabilities in**, you can land those costumers by **leveraging your platform's broader strengths**.

ELNs already integrating OpenCloning

eLabFTW

openBIS

Making strains or cell lines?

Support for strain and cell line design

- OpenCloning **tracks the genotype** of your strains and cell lines. **How they were made, which were their parents.**
- It supports **genome engineering methods** like Homologous Recombination, CRISPR-HDR, Cre-Lox and recombinases.

Want to speak to a user?

- Go and chat with ChangeBio, they make a ton of strains and are here in person!

Change Bio

The manufacturing layer for designed proteins

Working on lab automation?

The only automatable DNA design tool

- OpenCloning is the **only DNA design tool that supports automation** via python or web API.
- It also has the **only open data model for DNA engineering workflows**.
- Design your processes in the web editor or programmatically, and seamlessly turn them into reusable robotic workflows.

A DNA synthesis company?

Level up your web interface

- **No more copy-pasting** sequences in html forms.
- Provide your customers with a **rich web interface** to **design their constructs**.
- Allow them to **choose from your catalog of parts and backbones**.
- Users can **directly load genome sequences from the NCBI**, and run codon optimization.

Get started today!

Learn more

- Landing page - https://opencloning.org.
- GitHub - https://github.com/OpenCloning.
- Join the mailing list - Scan the QR on top.

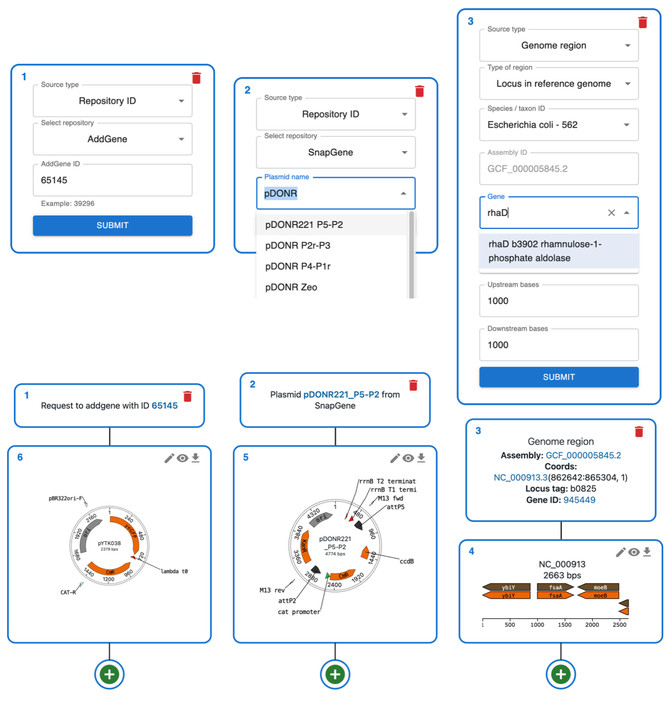
Hire support or custom development

- Reach out to manu@leraram.com.

Features

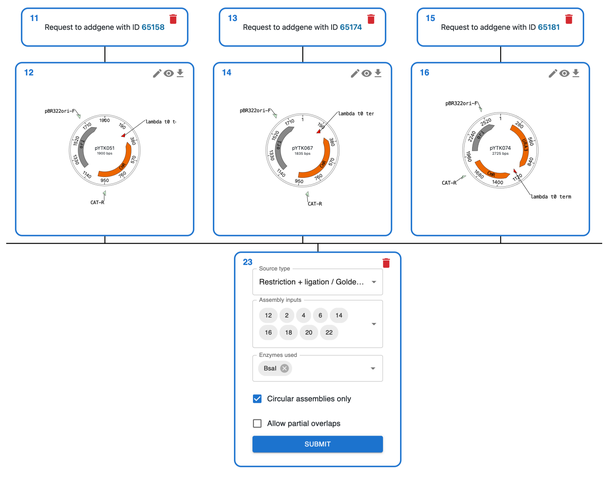
Import from anywhere

- Local files
- AddGene
- GenBank
- NCBI genomes
- SnapGene website
- Benchling link
- EuroScarf
- ELNs



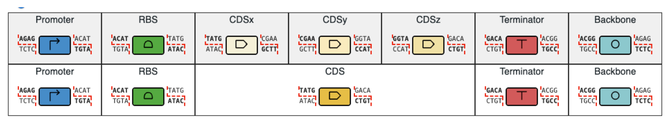
Techniques supported

- Golden Gate
- Gibson Assembly
- Restriction - ligation
- PCR
- Hom. Recombination
- CRISPR-HDR
- Overlap extension PCR
- Gateway™
- Polymerase extension
- Oligo hybridization
- *In vivo* assembly
- Cre-loxP
- Recombinases



Best Golden Gate tool

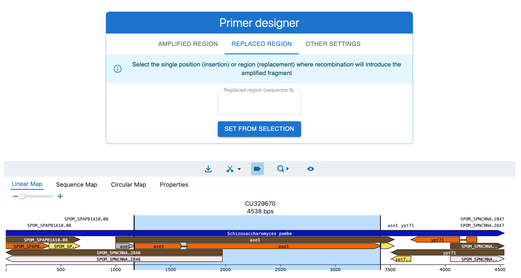
- **Define your syntax** in a simple user interface.
- **Plasmids get auto-assigned** to the right category.
- The **quickest and easiest way to do combinatorial assembly**.



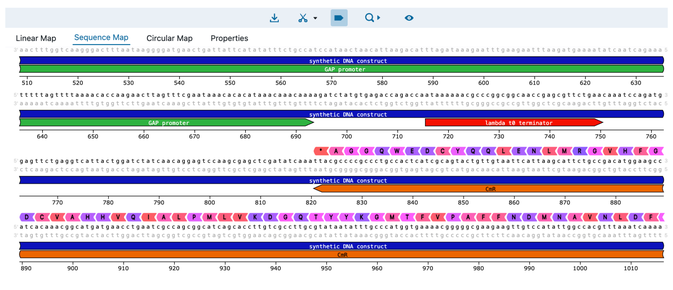
It's very easy to create your own syntax at https://syntax.opencloning.org/

Primer design

- Gibson Assembly
- Hom. Recombination
- CRISPR-HDR
- Restriction - ligation
- Checking PCR
- Gateway™



Rich feature viewer



Coming soon

- **Private sequence database for your lab**
  - Ancestry and progeny of all resources.
  - Rich queries (e.g., alleles of a certain gene, plasmids with a given marker)
- **Strain and cell line support**
  - Genotypes as arrays of alleles.
  - Alleles as cloning strategies starting from wt locus.
- **Managed hosting**
  - Host your sequences and designs in our cloud.
  - Access them from anywhere.